

AMPV V2 Supplementary Material

Table S1. Complete primary comet-like cell-resolved numerical summaries

C	K_{ref}	Workflow	Median $ e $	Vuln. (%)	Median signed e	Median realized κ
0	1	A	.048	10.3	.0061	7.024
0	1	B	.059	12.0	.0050	7.024
0	1	C	.255	36.0	-.0015	7.024
0	1	D	.060	12.3	-.0011	7.024
0	1	E	.012	4.3	8.972e-4	7.024
0	2	A	.024	6.3	.0031	14.047
0	2	B	.029	6.3	.0025	14.047
0	2	C	.125	20.7	.0015	14.047
0	2	D	.030	8.0	-.0014	14.047
0	2	E	.0060	1.7	-1.792e-4	14.047
0	5	A	.0096	2.7	.0012	35.119
0	5	B	.012	3.0	.0010	35.119
0	5	C	.049	10.0	-.0031	35.119
0	5	D	.013	3.3	-.0017	35.119
0	5	E	.0024	.3	-2.627e-4	35.119
0	10	A	.0048	.7	6.141e-4	70.237
0	10	B	.0058	1.0	5.025e-4	70.237
0	10	C	.026	7.3	-.0033	70.237
0	10	D	.0064	1.0	-.0017	70.237
0	10	E	.0012	0	-1.551e-4	70.237
0	20	A	.0024	0	3.071e-4	140.474
0	20	B	.0029	0	2.513e-4	140.474
0	20	C	.012	3.3	-.0041	140.474
0	20	D	.0038	0	-.0016	140.474
0	20	E	6.177e-4	0	-7.780e-5	140.474
0.6	1	A	.072	13.3	.065	7.024
0.6	1	B	.062	13.7	.038	7.024
0.6	1	C	.265	34.3	.016	7.024
0.6	1	D	.057	12.3	.037	7.024
0.6	1	E	.014	5.7	.0070	7.024
0.6	2	A	.037	6.0	.033	14.047
0.6	2	B	.031	6.0	.019	14.047
0.6	2	C	.130	20.7	.010	14.047
0.6	2	D	.029	7.3	.018	14.047
0.6	2	E	.0066	3.0	.0032	14.047
0.6	5	A	.015	3.0	.013	35.119
0.6	5	B	.013	3.3	.0078	35.119
0.6	5	C	.052	11.0	.0012	35.119
0.6	5	D	.011	3.0	.0056	35.119
0.6	5	E	.0027	.7	.0012	35.119
0.6	10	A	.0074	1.3	.0067	70.237
0.6	10	B	.0063	1.0	.0039	70.237
0.6	10	C	.027	6.0	-7.745e-5	70.237
0.6	10	D	.0053	.7	.0017	70.237
0.6	10	E	.0013	0	5.867e-4	70.237
0.6	20	A	.0037	.3	.0033	140.474
0.6	20	B	.0032	.3	.0020	140.474
0.6	20	C	.012	4.3	-7.970e-4	140.474
0.6	20	D	.0032	.3	-2.582e-4	140.474
0.6	20	E	6.540e-4	0	2.940e-4	140.474
1	1	A	.118	19.3	.114	7.024
1	1	B	.070	14.3	.059	7.024
1	1	C	.286	36.0	.047	7.024
1	1	D	.067	14.0	.055	7.024
1	1	E	.015	6.3	.011	7.024
1	2	A	.060	11.7	.059	14.047
1	2	B	.036	7.3	.030	14.047
1	2	C	.137	19.3	.026	14.047
1	2	D	.034	8.0	.027	14.047
1	2	E	.0075	3.7	.0054	14.047
1	5	A	.025	4.7	.024	35.119
1	5	B	.014	3.0	.012	35.119
1	5	C	.057	9.3	.010	35.119
1	5	D	.013	3.3	.0094	35.119
1	5	E	.0030	1.0	.0021	35.119
1	10	A	.012	2.3	.012	70.237
1	10	B	.0072	1.0	.0061	70.237
1	10	C	.027	7.3	.0036	70.237
1	10	D	.0061	1.0	.0035	70.237
1	10	E	.0015	0	9.953e-4	70.237
1	20	A	.0062	.3	.0060	140.474
1	20	B	.0036	.3	.0030	140.474
1	20	C	.014	4.0	4.754e-4	140.474
1	20	D	.0028	.3	7.874e-4	140.474
1	20	E	7.387e-4	0	5.012e-4	140.474

Table S2. Lowest tested reference contrast with vulnerability $\leq 5\%$

Workflow	$C = 0$	$C = 0.6$	$C = 1$
A	5	5	5
B	5	5	5
C	20	20	20
D	5	5	5
E	1	2	2

The threshold summary is descriptive and valid only on the tested grid.

Table S3. Compact-PSF inferential summary

All 40 contamination-versus-clean comparisons had 95% sample-identity bootstrap intervals above zero. The 16 clean-domain adjacent-reference-contrast comparisons were numerical-floor cases; all 32 contaminated-domain adjacent comparisons had intervals below zero. Across the 10 contaminated cells, A–E, B–E, and C–E were above zero in 10/10 cells for each comparison; A–B and A–C were below zero in 10/10; B–C included zero in 10/10. These results are task-family specific and do not establish a universal workflow ordering.

Table S4. SEP transfer cell summaries

κ	C	median $ e $	95% CI	observed vuln.
1	0	3.767e-09	[2.655e-09,4.879e-09]	0.00
1	0.6	0.169792	[0.101714,0.199482]	0.01
1	1	0.282987	[0.172175,0.332470]	0.10
5	0	3.780e-09	[2.908e-09,4.977e-09]	0.00
5	0.6	0.0339584	[0.0206610,0.0398964]	0.00
5	1	0.0565974	[0.0344350,0.0664940]	0.00
20	0	3.780e-09	[2.957e-09,4.977e-09]	0.00
20	0.6	0.00848961	[0.00508637,0.00997500]	0.00
20	1	0.0141493	[0.00860874,0.0167451]	0.00

Observed vulnerability fractions use $\tau = 0.5$; no AE04 vulnerability confidence interval is used.

Table S5. SEP contamination contrasts

κ	C	median difference	95% CI
1	0.6	0.169792	[0.101727,0.199482]
1	1	0.282987	[0.172175,0.334901]
5	0.6	0.0339584	[0.0203455,0.0400423]
5	1	0.0565974	[0.0344218,0.0667371]
20	0.6	0.00848961	[0.00516524,0.00979792]
20	1	0.0141493	[0.00860874,0.0163907]

All six intervals are above zero.

Table S6. SEP reference-contrast contrasts

C	κ low–high	median difference	95% CI
0	1–5	3.874e-10	[-7.080e-10,1.471e-09]
0	5–20	0	[0,0]
0.6	1–5	-0.135834	[-0.156767,-0.0826439]
0.6	5–20	-0.0254688	[-0.0299223,-0.0154957]
1	1–5	-0.226390	[-0.265976,-0.137740]
1	5–20	-0.0424480	[-0.0498750,-0.0254319]

The four contaminated-domain intervals are below zero. The two clean-domain comparisons are numerical-floor cases and are not counted as directional transfer evidence.

Scope

The SEP experiment is an implementation-transfer test on synthetic compact-PSF scenes with known source position. It is not observational validation, does not establish SEP superiority, and does not extend the tested reliability domain beyond the stated grid.